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Studierichting: Informatica
Oefeningenreeks 7C nr. 1.2

$$f(x,y,z,t) = \ln \frac{xy}{zt}$$

Eerste orde:

$$\frac{\partial f}{\partial x} = \frac{zt}{xy} \cdot \frac{y}{zt} = \frac{1}{x}$$
$$\frac{\partial f}{\partial y} = \frac{zt}{xy} \cdot \frac{x}{zt} = \frac{1}{y}$$
$$\frac{\partial f}{\partial z} = \frac{zt}{xy} \cdot \left(-\frac{xy}{z^2 t}\right) = -\frac{1}{z}$$
$$\frac{\partial f}{\partial t} = \frac{zt}{xy} \cdot \left(-\frac{xy}{zt^2}\right) = -\frac{1}{t}$$

Tweede orde:

$$\frac{\partial^2 f}{\partial x^2} = -\frac{1}{x^2}$$
$$\frac{\partial^2 f}{\partial y^2} = -\frac{1}{y^2}$$
$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{z^2}$$
$$\frac{\partial^2 f}{\partial t^2} = \frac{1}{t^2}$$

$$\Rightarrow D^2 f(x,y,z,t) = \begin{pmatrix} -\frac{1}{x^2} & 0 & 0 & 0 \\ 0 & -\frac{1}{y^2} & 0 & 0 \\ 0 & 0 & \frac{1}{z^2} & 0 \\ 0 & 0 & 0 & \frac{1}{t^2} \end{pmatrix}$$

References